

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every 7 Months

Thomas Kwa, Ben West, Joel Becker, et al. —
METR

Source: <https://arxiv.org/abs/2503.14499>

A New Metric: 50% Task-Completion Time Horizon

Methodology Flow:

1. Assemble diverse suite of 170 software/research tasks.
2. Measure human timing and AI success rates.
3. Fit logistic model to find the duration at which AI succeeds 50% of the time.

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: “Which file is a shell script?” Choices: “run.sh”, “run.txt”, “run.py”, “run.md”
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

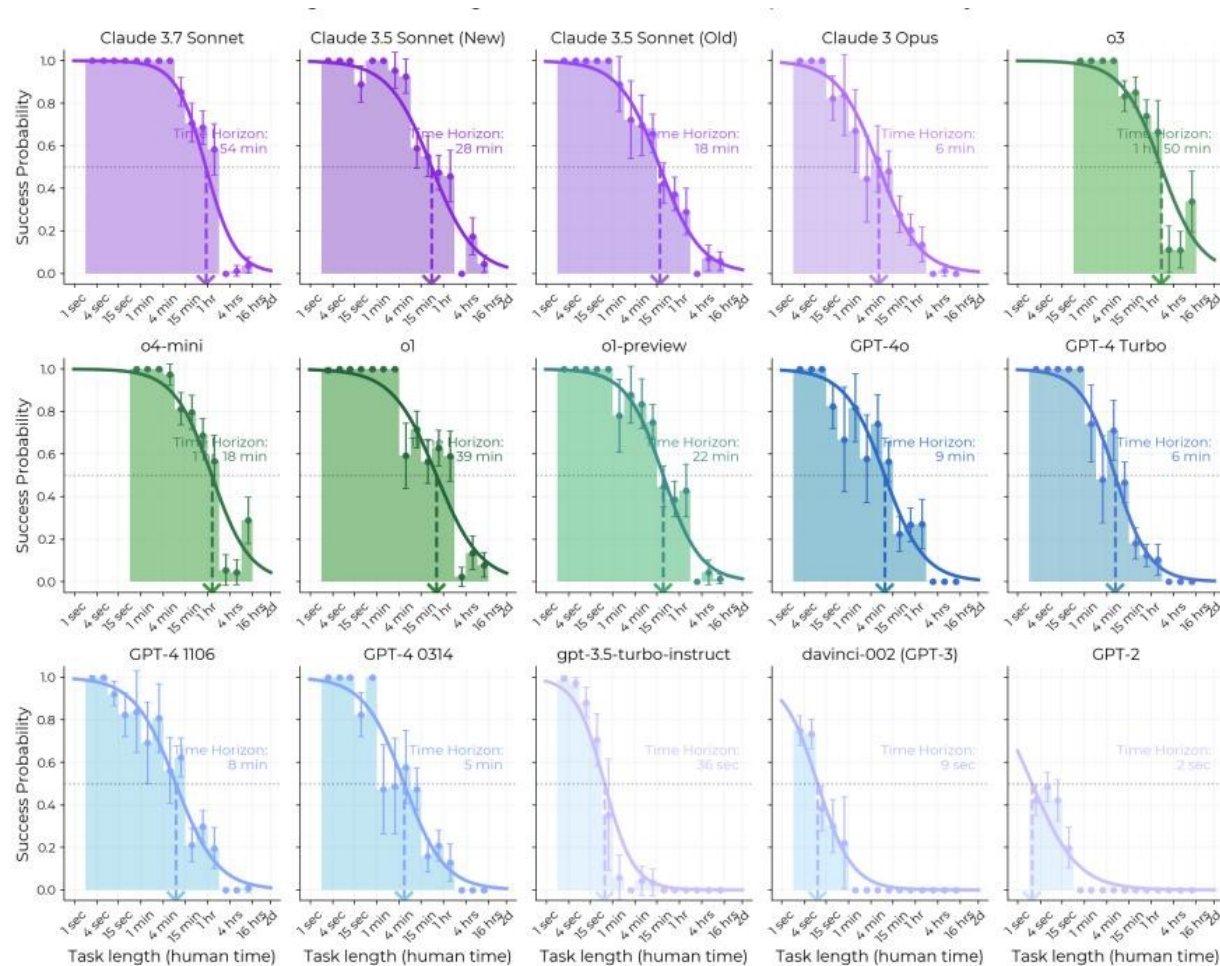
Suite Composition: HCAST (97 tasks), RE-Bench (7 tasks), and SWAA (66 tasks).

Family	Human Time	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?"
wikipedia_research	1 minute	Research simple factual information from Wikipedia
oxdna_simple	9 minutes	Detect and fix a bug in molecular dynamics simulation input files
munge_data	56 minutes	Write a Python script to transform JSON data by inferring conversion rules
cuda_backtesting	8 hours	Implement custom CUDA kernels for 30x speedup of a backtesting tool

AI Task Horizon Grew from 2 Sec to ~1h

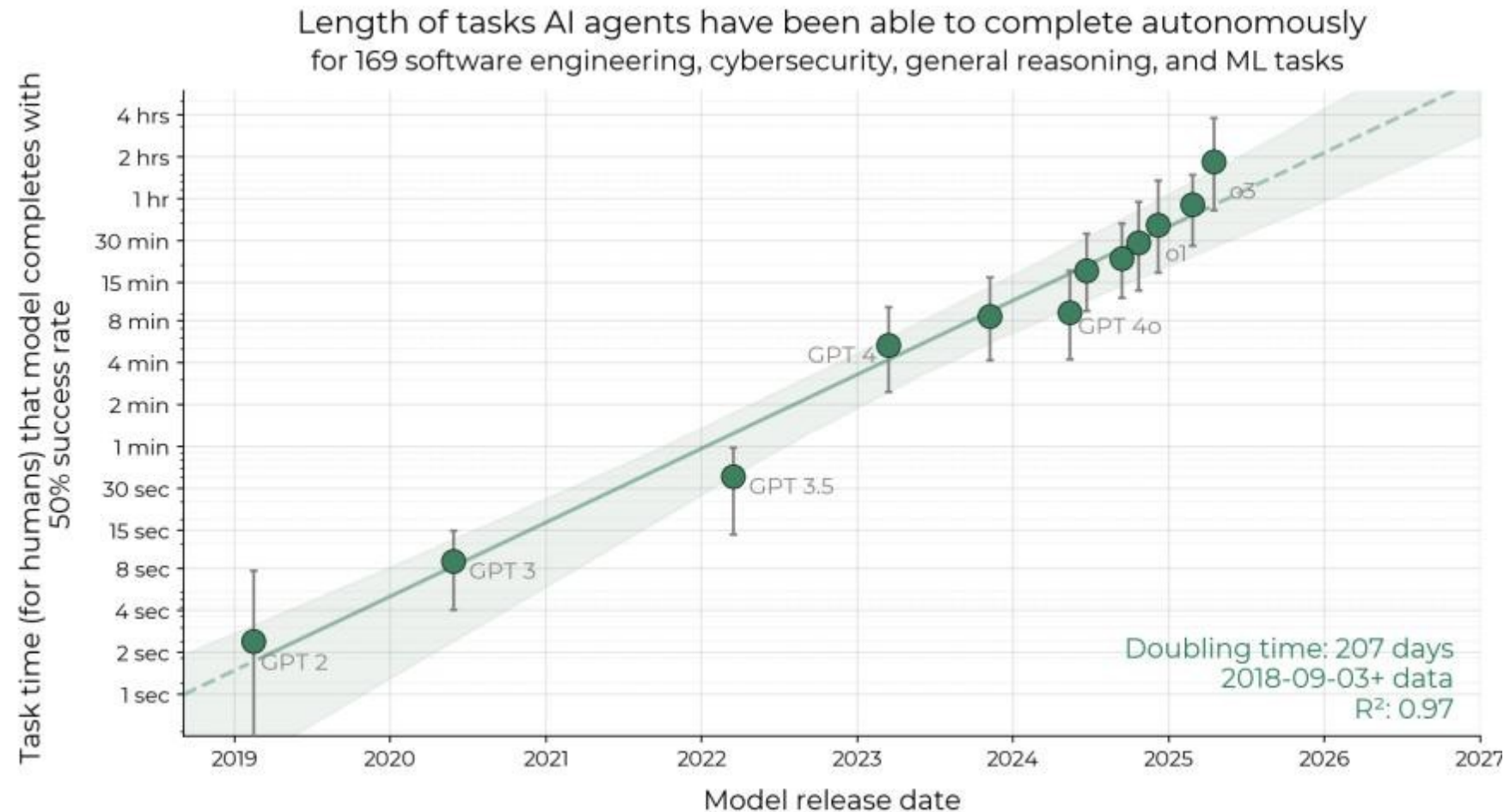
50m in 6 Years

Key Findings: ~207-day doubling time (~7 months), $R^2=0.97$. GPT-2 (~2s) -> GPT-4 (~5m) -> o3 (~1h 50m).



Success Probability Drops Sharply as Task Length Increases

Frontier models (Claude 3.7 Sonnet, o3) push the curve rightward compared to older models like GPT-2.



Reasoning, Tool Use, and Error Recovery Drive Growth

- **Key Capability Drivers:**
 - Improved logical reasoning.
 - Better tool use and environment interaction.
 - Greater reliability and self-awareness in task execution.
 - Ability to adapt to mistakes and recover from errors.
- **Notable Limitations:**
 - Performance is significantly lower on unstructured, 'messier' tasks.
 - External validity remains an open question—tasks may not perfectly map to typical intellectual labor.

Extrapolation Predicts Month-Long Automation by 2028-2031

- **Future Projections & Caveats:**
 - Naive extrapolation predicts AI reaches a 1-month time horizon (167 work hours) between mid-2028 and mid-2031.
 - Implies potential automation of many month-long software projects and related tasks.
 - Recent acceleration: 2023–2025 growth rate is roughly 20% faster than the 2019–2025 average.
 - Caveats regarding the forecast:
 - Trends may saturate due to structural bottlenecks or data walls.
 - Performance on benchmarks may decouple from real-world economic utility.

Takeaways: Exponential Growth Demands Proactive Governance

- **Summary of Key Findings:**

- 50% time horizon is a robust, intuitive metric for tracking AI autonomy ($R^2 = 0.97$ over 6 years).
- Time horizon is doubling every ~7 months, with potential acceleration since 2024.
- Current frontier is ~2 hours of human-equivalent work (model o3).
- External validity is an open question: tasks may not perfectly represent real-world intellectual labor distributions.
- Capabilities excel in structured logic but face limitations in highly unstructured domains.